

A black and white photograph of Nancy Cartwright, a philosopher of science. She is wearing glasses, a white t-shirt, and a patterned scarf. She is positioned in front of a bookshelf filled with books. The word "UNDERDETERMINATION" is overlaid in large blue capital letters on a white rectangular background.

UNDERDETERMINATION

Nancy Cartwright (1944-)
Philosopher of Science

ANNOUNCEMENT

Assignment: Role Play of a Scientific Figure

Deliverable 2: Script – due Feb 6th, 11:59 pm

Deliverable 3: Full Submission – due Feb 15th, 11:59 pm

ANNOUNCEMENT

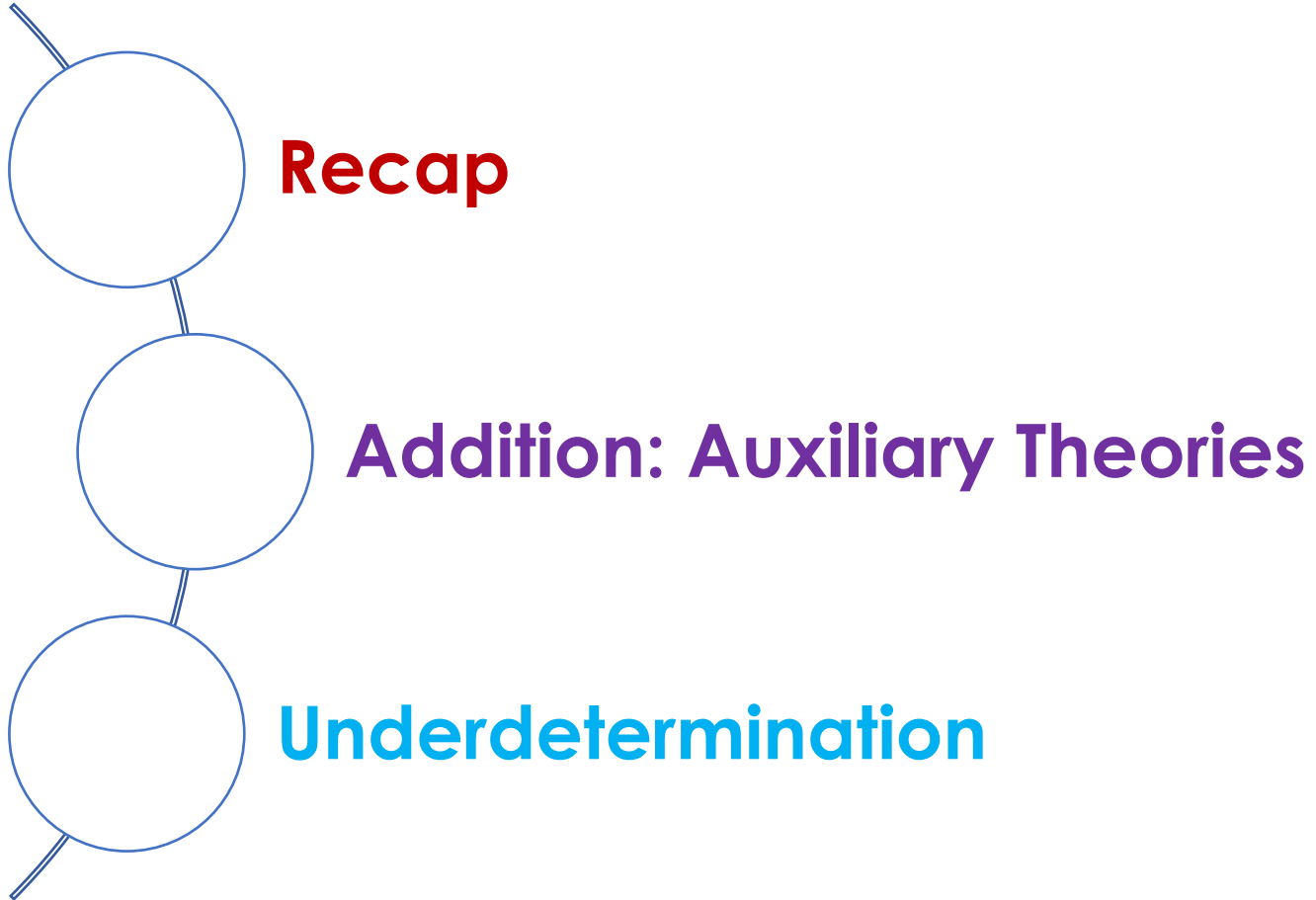
Assignment: Role Play of a Scientific Figure

Deliverable 2: Script – due Feb 6th, 11:59 pm

Please incorporate some of the concepts we have discussed in our philosophy of science module.

I will be looking for at least **1 to 2 examples** connecting what we have learned to the work and legacy of the scientific figure you have chosen.

The Plan!

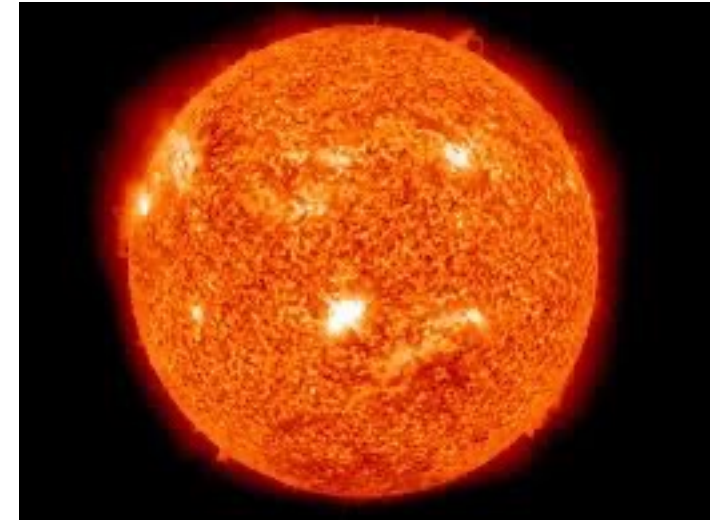


RECAP

CONFIRMATION



- We find many *competing* claims about unseen events and objects that explain the phenomena we see:
 - Example: Where does the Sun's energy come from?
 - From the Greek deity Helios.
 - The Sun is a burning ball of coal or fuel.
 - The Sun is slowly shrinking due to gravity, converting gravitational energy into heat and light.
 - Radioactivity is the Sun's energy source.
 - The Sun's energy source is an ongoing process of nuclear fusion, which showers us with heat, light, and neutrinos.



But at most one of those theories is true (*likely to be true*).

CONFIRMATION



- The responsibility of science is to weed out the 'bad' theories from the 'good'.
- **How?**
 - By testing the theory against the observable data.

CONFIRMATION



- The responsibility of science is to weed out the 'bad' theories from the good.
- **How?**
 - By testing the theory against the observable data.



But what do we mean by
“testing” a theory and “by
comparison with the
observable evidence”?

CONFIRMATION



Example:

- Theoretical claim: “Helium atoms contain two electrons.”
- Checking is not a matter of looking around and counting (we cannot directly observe these things!).
- Rather, it is a matter of *checking on the consequences* of the theoretical claim by determining what *effect* the electrons must have on the *observable world*.

Last time!

INTRODUCTION

When we “reduce” electron to its experiential consequences, we get something like bubbles in a cloud chamber. We don’t see electrons directly. We see the bubble trail it leaves.



Cloud Chamber

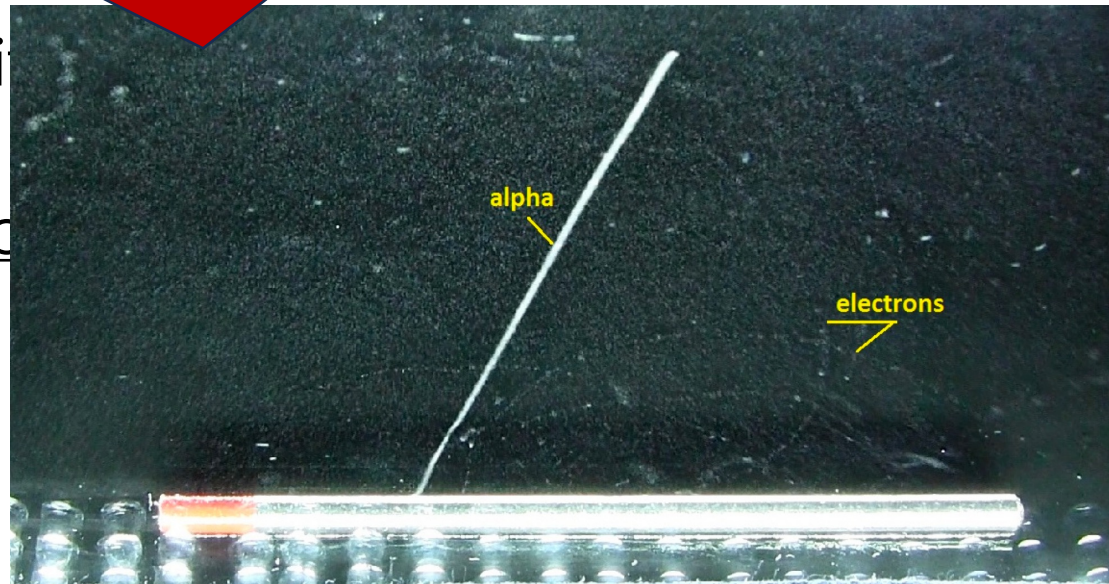
heat, etc. The phenomena is observable.

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There is a kind of **indirectness** in any case of using observational evidence to test a theory.



CONFIRMATION



- However,
 - A theory *must* be accountable to *observational justification*.
 - This means that not only must a theory be **testable** (internal virtue), but it must be also be **tested**.

Last time!

CONFIRMATION

- Example: Where does the Sun's energy come from?
 - The Sun's energy source is an ongoing process of nuclear fusion, which showers us with heat, light, and neutrinos.
 - And some of the neutrinos ought to find their way to the earth where they can be detected.
 - The theory is testable since the neutrinos are detectable.



Raymond David Jr. detected solar neutrinos by using a 100,000-gallon tank of chlorine-rich cleaning fluid placed 1.5 km underground in a gold mine. Neutrinos occasionally interact with the chlorine, changing it into a radioactive argon atom. The radioactive emissions from the argon can then be counted as the footprints of solar neutrinos.



CONFIRMATION

- Confirmation and explanation follow the *same logical form*:
 - Both *compare the consequences of a theory to observations in the world* and thus connect theories with the world by observing the effects that the theorized entities have on objects we can see.
 - Example: *Solar Theory*
 - If the neutrinos are detected we will say that they contribute to the *confirmation* of the solar theory.
 - And that the solar theory *explains* the shower of neutrinos.

CONFIRMATION



- It is typically, though,
 - to apply the concept of ***explanation*** to ***past observations*** (things we knew happened but wondered why),
 - and reserve the ***concept of testing*** to cover ***future observations***, predictions whose outcome is yet unknown.

CONFIRMATION



Last time!

- Explanations and confirmation are also distinguished by the pragmatics of the situation:
 - If it is the phenomenon itself that is puzzling and the question is “***Why does this happen?***” then it is ***explanation*** scientists are after.
 - If the interest is centered on the ***status of the theory itself*** and the phenomena are just hoops it must jump, then it is ***confirmation***.

A BASIC MODEL OF CONFIRMATION

Last time!

Hypothetico-deductive confirmation (H-D):

- This is a very general, formal model of confirmation in which a *hypothesis* is tested by its *deductive consequences*.
- It is by no means a deductive proof that the hypothesis is true.
- Rather, it is a way of providing *inductive* evidence that the hypothesis is *likely to be true*.

For **example**, the hypothesis that all dogs are vicious has the deductive consequence that the collie next door is vicious. I test this by testing the disposition of the collie next door. If she is vicious, then this is **some small** indication that the hypothesis is true.



A BASIC MODEL OF CONFIRMATION



- If one successful H-D test helps a little, further H-D tests will help more.

Example:

- The theory of relativity does not rest its credibility on the single success of predicting the bending of starlight.
- It makes other predictions, for example, that gravity affects the passage of time and the frequency of light. This is the gravitational red shift, and it too has been observed (e.g., clocks on Earth tick more slowly than those in satellites).

A BASIC MODEL OF CONFIRMATION



- In the case, where the prediction fails, we speak of ***hypothetico-deductive falsification***:
 - The idea is that if the theorized object or event is *always* supposed to cause the observable effect mentioned, but in fact the effect is missing, then the theorized object must not be there.

A BASIC MODEL OF CONFIRMATION



- Note, there is an ***asymmetry*** in this model of testing between confirmation and falsification.
 - ***Confirmation*** comes *in degrees* and is vulnerable to confusion as to how much is enough.
 - ***Falsification***, on the other hand, shows no such sloppiness, no degrees, no need for discretion.
 - A single failed prediction will suffice to falsify a theory and force its rejection.

• The solar-neutrino problem is not a big enough problem to force the rejection of the theory.

In other words, the failure to observe the predicted effect has not been taken as conclusive proof that the theory is false.

detectable.

- **Problem:** Discrepancy between the number of predicted neutrinos and the number Raymond David Jr. measured (known as ***Solar-Neutrino Problem***)



Last time!

Raymond David Jr. detected solar neutrinos by using a 100,000-gallon tank of chlorine-rich cleaning fluid placed 1.5 km underground in a gold mine. Neutrinos occasionally interact with the chlorine, changing it into a radioactive argon atom. The radioactive emissions from the argon can then be counted as the footprints of solar neutrinos.

• The solar-neutrino problem is not a big enough problem to force the rejection of the theory.

In other words, the predicted value is taken as conclusive evidence that the theory is false.

It also doesn't take the experimental setup (e.g., the fluid may be contaminated) and the reasoning process that issued the prediction into account.

Discrepancy between the number of detected neutrinos and the number Raymond David Jr. measured (known as **Solar-Neutrino Problem**)



Raymond David Jr. detected solar neutrinos by using a 100,000-gallon tank of chlorine-rich cleaning fluid placed 1.5 km underground in a gold mine. Neutrinos occasionally interact with the chlorine, changing it into a radioactive argon atom. The radioactive emissions from the argon can then be counted as the footprints of solar neutrinos.

Last time:

A BASIC MODEL OF CONFIRMATION



- So it is more complicated as the simple H-D model indicates, ...
 - (e.g., a two-component deduction from the hypothesis about solar fusion to the prediction of clicks on the detector counting radioactive argon.)
- ... and a realistic model of scientific testing must take the details seriously.
 - (e.g., the conditions of testing.)

MORE DETAILED & REALISTIC MODEL OF CONFIRMATION



Last time!

- The *three components* of the revised ***hypothetico-deductive model*** are
 1. the hypothesis (that is, the theory being tested),
 2. the conditions of testing, and
 3. the predicted effect.
- If the hypothesis is true, then if conditions are right, the effect will be observed.
- If, in fact, the effect is not observed, then we are forced to conclude that ***either*** the hypothesis is false ***or*** the conditions are not right.

MORE DETAILED & REALISTIC MODEL OF CONFIRMATION



- In addition, it becomes clear that more than one theoretical claim is needed to deduce the predictions.
- The hypothesis alone does not entail the prediction.
- It is rather the hypothesis in conjunction with ***auxiliary theories***,
 - i.e., *the background knowledge required to draw testable consequences from a hypothesis*
- from which predictions are deduced.

MORE DETAILED & REALISTIC MODEL OF CONFIRMATION



- No hypothesis alone entails testable predictions, so auxiliary theories always participate in the testing of a hypothesis.
 - For example, the hypothesis that the solar interior is a furnace of nuclear fusion has the testable consequence that solar neutrinos are showering the earth, but the consequence can be drawn out only with the theoretical knowledge that fusion produces neutrinos. The claim about the production of neutrinos is an auxiliary theory in this case of testing.

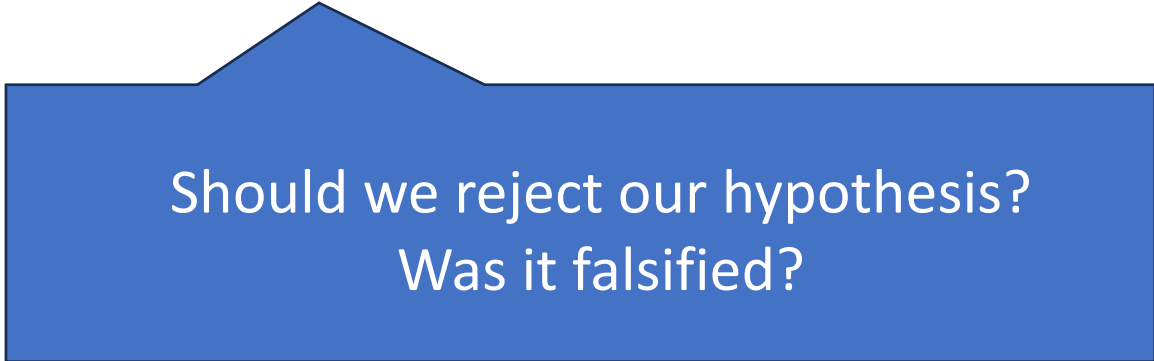
Addition: Auxiliary Theories

Addition: Auxiliary Theories

- An example:
 - In order to test one hypothesis, you have to make many assumptions
 - These assumptions are sometimes called “background conditions” or “auxiliary theories”

Addition: Auxiliary Theories

- A hypothesis: "High air pressure is associated with fair, stable weather."
 - We make a series of observations: we find that high air pressure is associated with unstable weather.



Should we reject our hypothesis?
Was it falsified?

Addition: Auxiliary Theories

- A hypothesis: "High air pressure is associated with stable weather."
 - We make a series of observations and find that high air pressure is associated with stable weather.

Not necessarily!

Should we reject our hypothesis?
Was it falsified?

Addition: Auxiliary Theories

- What we have found is that: the observations we have are in conflict with the predictions of our hypothesis.
 - **Quine-Duhem Thesis:** This might mean that the measuring instruments are faulty, or the circumstances of observation may be wrong, maybe observers are unreliable, or records are unreliable, or you've used the wrong computer program to analyze the data.
 - All these count as **background assumptions** or **auxiliary theories**. (They exist in the background whenever we test a certain hypothesis.)

Addition: Auxiliary



This is nothing new! You all already know this.

We often fail to get the right results in science labs in school, even for theories we all **know** are 'true'.

These results do not falsify the theories. We blame *ourselves*

Another Example: The OPERA Case



Addition: Auxiliary Theories

- Therefore:
 - When you test a single hypothesis, what you are really testing is a long conjunction of statements (hypothesis + all background assumptions).
 - If your testing gives unexpected results, then something in the conjunction is false, but this does not tell you where the error is!
 - The problem may lie in your hypothesis but it also may lie in your background assumptions.

Addition: Auxiliary Theories

- However, this also contains some problems:
 - If Quine is right, then no scientific hypothesis or statement is, in and of itself, falsifiable.
 - All can be saved from falsification by giving up background assumptions, or by making adjustments.

Addition: Auxiliary Theories

- However, this also contains some problems:
 - If Quine is right, then no scientific hypothesis or statement is, in and of itself, falsifiable.
 - All can be saved from falsification by giving up background assumptions, or by making adjustments.
- However, this doesn't mean that all theories are equally good.
 - The question then becomes this: On what grounds are such judgments made?

Underdetermination

Underdetermination



Reflect & Discuss

Read the first passage on your worksheet

1. *"observation of evidence cannot stand as clear confirmation of the theory."*

- **How can this be true given that external virtues are heavily weighted in the justification of a theory?**
- **How can evidence not be a decisive factor in deciding between theories?**

VIRTUES

EARLIER!

- **Epistemic Virtues (truth-conducive virtues) of Scientific Theories:**

- **Internal Virtues:**

- Internal features of theories that are truth-conducive

- **External Virtues:**

- External features that indicate that the theory is compatible with experimental results, that is, that the theory is accountable to observations.

External virtues do not determine
that a theory is true or false.

EARLIER!

- What does it mean for a theory to be explanatory (scientifically)?
 - Scientific explanations are used to explain the world
 - and are used to predict future events
 - Explanations are used to understand the world

However, consider the following examples:

The Miasma theory explains the cholera and the plague.

The theory of oceanic sirens explains the tides.

That explains phenomena is more likely to be true



A theory can be explanatory ...

AMENDMENTS TO THE COVERING-LAW MODEL



- A theory can be explanatory without being true.
 - Example:
 - The explanation of my neighbor stealing the newspaper might make sense, but it does not mean that it is true.
 - Nonetheless, if I continue to think as if the theory is true, I continue to make sense of the phenomena in my world. Now that I understand the phenomenon, I can deal with it (e.g., I know the newspaper will not be there. Thus, I need to go and buy a new one).
- **The explanation fully serves its purpose even though its theory is false.**

... and can be favorably tested against the observable evidence yet still be false.

Underdetermination

- The logical structure of an explanation and testing is ***never*** of the *form that the evidence entails a theory*: Rather, it is the other way around.
 - *A theory entails its evidence, and because of this, observation of the evidence cannot stand as clear confirmation of the theory.*
 - Nor can the evidence provide unambiguous falsification since, as we have seen (e.g., solar neutrino problem, OPERA case), *realistic cases of testing are of the form, theory **plus auxiliaries** entail the evidence.*

Underdetermination

To put it bluntly,

- the observations of the world, that is, the reports of *manifest* (observable) phenomena, are *not sufficiently informative* to single out the one true theory of what is going on behind the *appearances*.

Underdetermination

To put it bluntly,

- the observations of the world, that is, the reports of *manifest* (observable) phenomena, are *not sufficiently informative* to single out the one true theory of what is going on behind the *appearances*.
 - **Note:** Explanation and testing are strict and crucial discriminators of scientific theories in that a theory is unacceptable if it is in *no way explanatory* and *does not cooperate with other theories to make true predictions*.
 - The comparison of a theory with observations weeds out lots of potential theories ...

Underdetermination

- The problem of **underdetermination** is that this process doesn't weed out enough, since many competing theories can account equally well for the evidence.
- *The evidence alone cannot trim the pool of applicants down to the single theory that is the true theory.*

Underdetermination

- As long as an alternative account is conceptually possible (and it always is), the observational data alone can give the theory we have in mind at best a fifty-fifty chance of being true.
- In fact, though, the **problem of underdetermination** is much worse than this because,
 - for any finite amount of data (things explained, tests passed), there will always be *any number* of theories that are empirically equivalent over that data.

Underdetermination

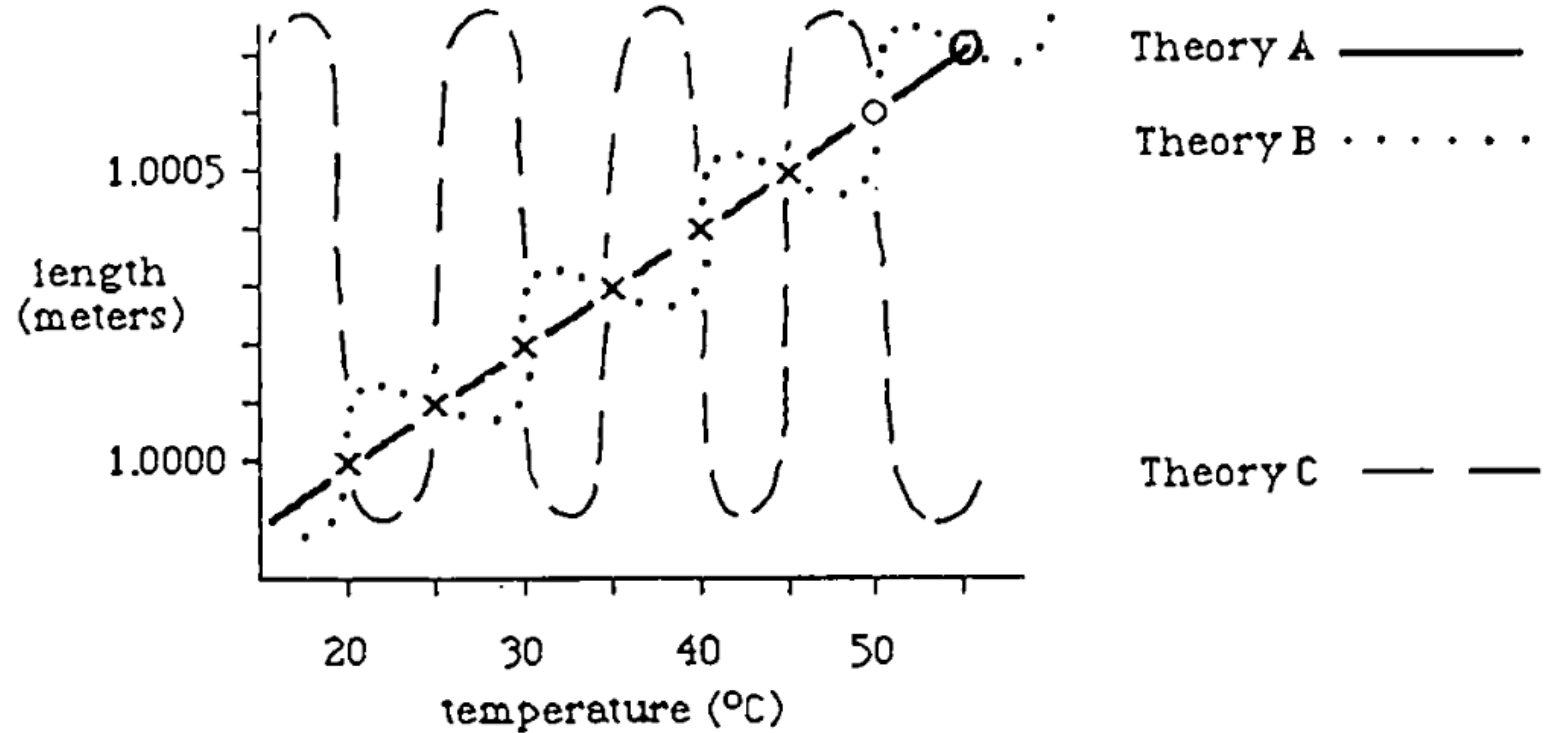
- The pool of empirically adequate theories can be trimmed toward the truth only in cooperation with *internal standards (virtues)*.
 - Internal virtues:
 1. Entrenchment
 2. Explanatory cooperation
 3. Testability
 4. Generality
 5. Simplicity



Underdetermination

The heated-bar example

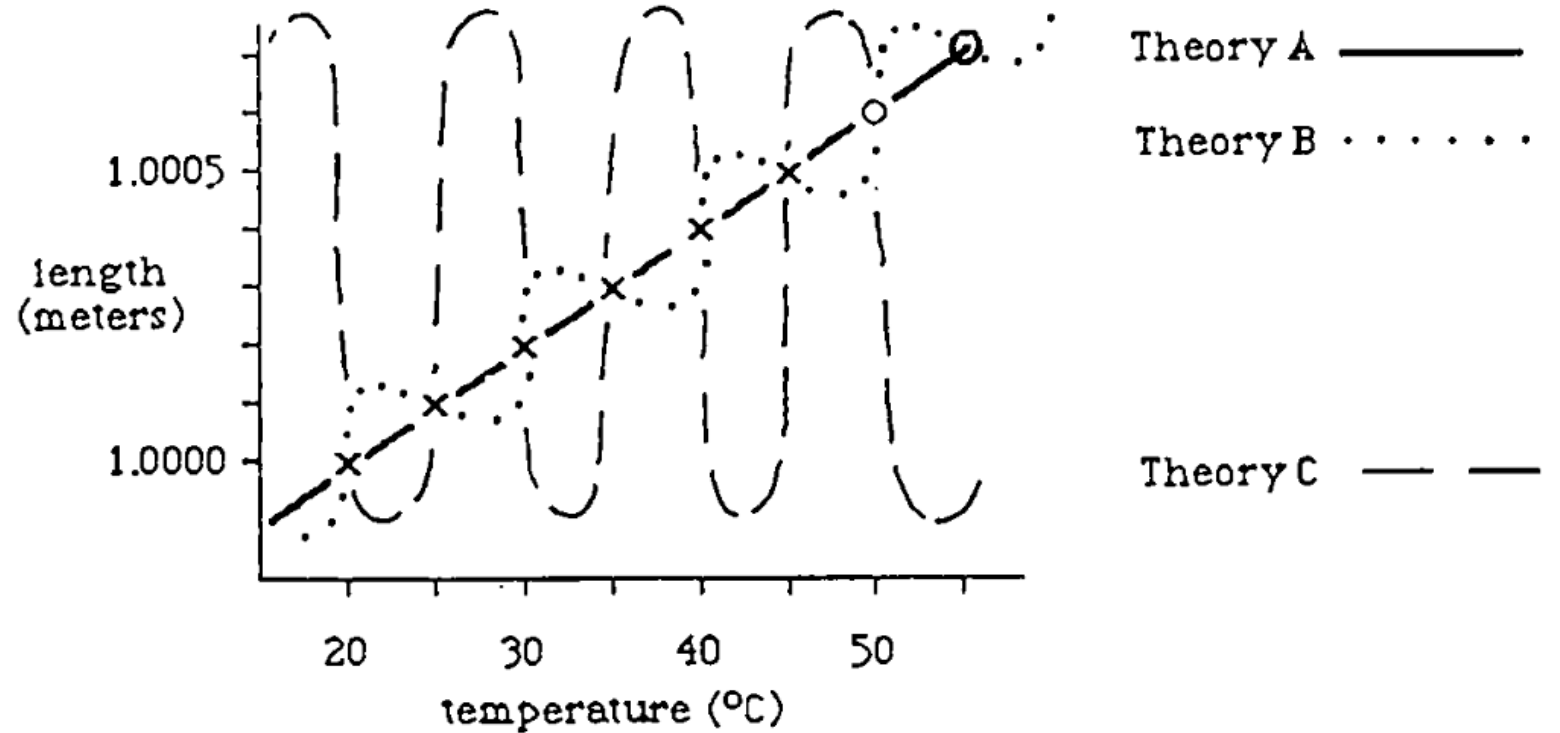
A theory of what is going on here, that is, of the mechanism by which heating the bar causes it to change size, will describe what length the bar *would* take at any particular temperature.



Underdetermination

The heated-bar example

Theory A explains all the observed events and correctly predicts the future test at 50 degrees.

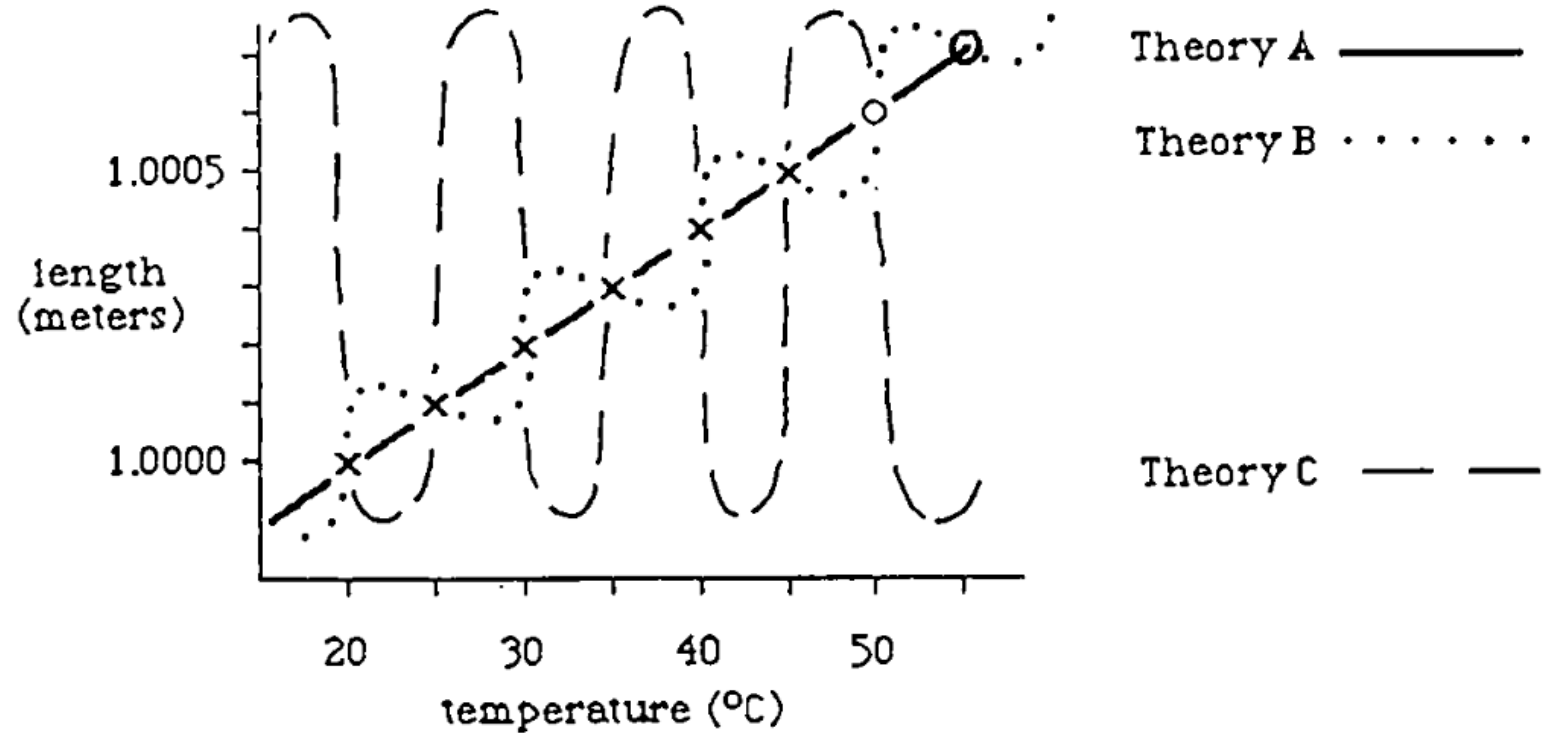


Underdetermination

The heated-bar example

Theory A explains all the observed events and correctly predicts the future test at 50 degrees.

But so does theory B.



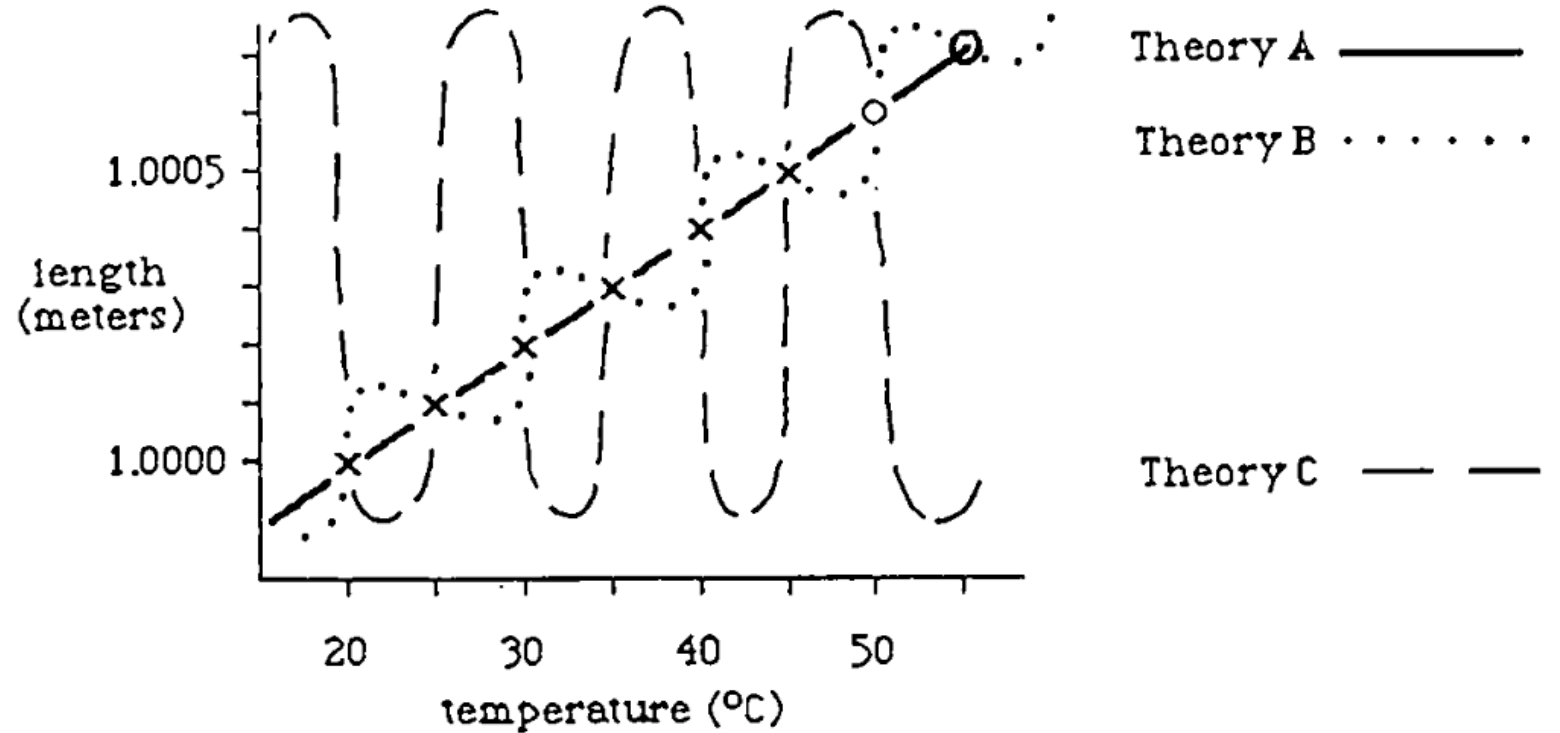
Underdetermination

The heated-bar example

Theory A explains all the observed events and correctly predicts the future test at 50 degrees.

But so does theory B.

Theory A is preferable on the standards of *simplicity*.

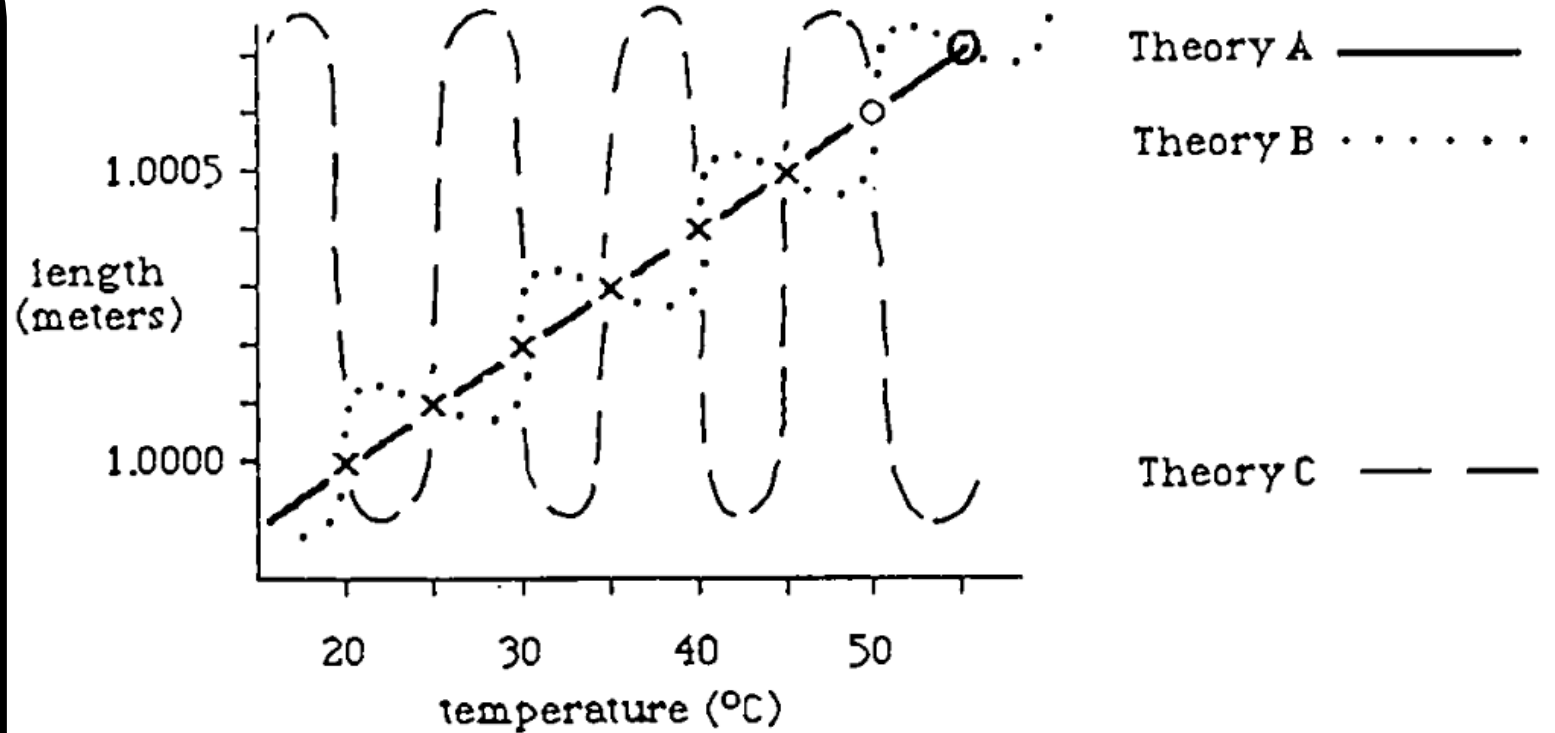


Underdetermination

The example demonstrates that the world *seems* as if theory A is true.

But based on the observation, it appears as well as if B is true, and clearly both cannot be right.

This forces a close look at the question of the inference from evidence to theory, the inference from “seems as if” to “is.”



Underdetermination



Reflect & Discuss

2. The philosophical stances taken in response to the problem of underdetermination weaken the commitment of science to truth.

- **Name and describe these stances**

Underdetermination

1. Instrumentalism:

- This brand of antirealism claims that scientific theories are not intended to be evaluated as being true or false.
 - Theories are *useful* or not, *applicable* or not, but they are not *objectively* true or false.
 - They are *instruments* or tools like a hammer, which is judged for its usefulness and its applicability to a particular task (e.g., for driving nails).



Underdetermination

1. Instrumentalism:

- This brand of antirealism claims that scientific theories are not intended to be evaluated as being true or false.
 - Theories are useful or not, applicable or not, but they are not *objectively* true or false.
- They are only ***instruments*** we use to guide our thoughts and beliefs about the ***world we can observe***.
 - Theories are intellectual tools for purpose of prediction of future phenomena and organization of observations.

Underdetermination

2. The Position of the Empiricist:

- This is a kind of antirealism that claims that scientific theories are in fact true or false in what they say about both observables and unobservables, but we can have justified knowledge only about what they say about observables.

Underdetermination

2. The Position of the Empiricist:

- This is a kind of antirealism that claims that scientific theories are in fact true or false in what they say about both observables and unobservables, but we can have justified knowledge only about what they say about observables.
- The responsible position to take with regard to claims about what cannot be observed is a position of agnosticism: Withhold belief in those claims that are beyond the limits of knowledge.

Underdetermination

3. Realism:

- assumes that the theories of science are in fact true or false of the world, or at least approximately so, and that we can sometime tell of a theory whether it is true or false.

Underdetermination

3. Realism:

- assumes that the theories of science are in fact true or false of the world, or at least approximately so, and that we can sometime tell of a theory whether it is true or false.
- In other words, science can and does make justified knowledge claims about aspects of the world that cannot be observed.

Underdetermination

- Let's take a close look at the different philosophical stances in response to the problem of underdetermination ...

Underdetermination

1. Instrumentalism:

- evaluates scientific theories in terms of *usefulness* and *applicability*, rather than in terms of truth.
- *Empirical adequacy* – efficient organization and successful prediction of phenomena – is the goal of theorizing.

Underdetermination

1. Instrumentalism:

- evaluates scientific theories in terms of *usefulness* and *applicability*, rather than in terms of truth.
- *Empirical adequacy* – efficient organization and successful prediction of phenomena – is the goal of theorizing.
 - Equally empirical adequate theories are equally acceptable. As there is no one true hammer, there is no one true theory of gravity.

Underdetermination

1. Instrumentalism:

- On this view, theories function as pedagogical tools for experts.
 - For example, science teachers might use a conceptual aid to help their students understand the physical world. The atoms in a molecule, for instance, can be visualized as being connected by inner springs.

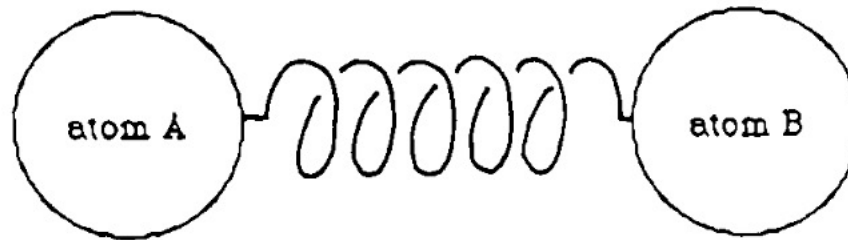


Figure 5.2. A model of the diatomic molecule AB , with the binding force represented as a spring.

1. Instr

- On t
-

This conceptualization of springs between atoms helps us to get the calculations right ...

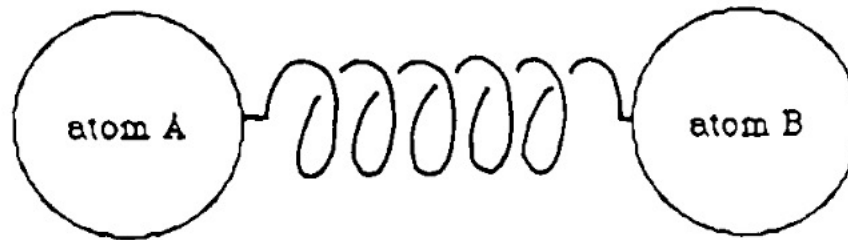


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This conceptualization of springs between atoms helps us to get the calculations right ...

1. Instr

- On t
-

But no science teacher means to say that there really *are* tiny springs between the atoms.

The claim about the real world is only that it is *as if* there are such springs.

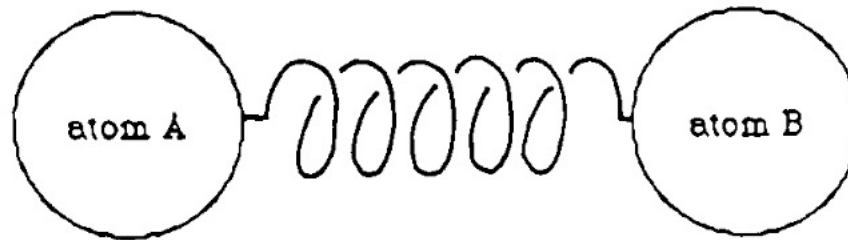


Figure 5.2. A model of the diatomic molecule AB , with the binding force represented as a spring.

Underdetermination

1. Instrumentalism:

- Similarly, unobservable entities (such as electrons, quarks, atoms) *should not* be taken *literally*, and so it would be inappropriate to call them true or to call them false.
- Thus, we should stop endorsing talking about unobservables like electrons and atoms as true.

Underdetermination



Reflect & Discuss

3. “Empirical adequacy – efficient organization and successful prediction of phenomena – is the goal of theorizing.”

- **Why might someone subscribe to this view?**
- **What doctrine opposes this view and how?**
- **How is the atheist similar to the instrumentalist?**

(religious atheism: i.e., either the lack of belief that there exists a God, or the belief that there exists none.)

Underdetermination

- One of the reasons why one might subscribe to the view of instrumentalism is to distance oneself from the mystical and the occult (e.g., unobservable forces as found in astrology).
 - There are no such things as electrons, quarks, or atoms, just as there is no such thing as God.

Underdetermination

- One of the reasons why one might subscribe to the view of instrumentalism is to distance oneself from the mystical and the occult (e.g., unobservable forces as found in astrology).
 - There are no such things as electrons, quarks, or atoms, just as there is no such thing as God.
- This happens for the price of really understanding what is happening behind the scenes in the realm of unobservables – **instrumentalism** ignores the scientific surprise and curiosity about the unobservable.
 - However, something *causes* the world as we see it, meaning, observable objects are constituted of something. And we want to know about it. This kind of knowledge, contrary to instrumentalism, is and should be the goal of theorizing.

Underdetermination



Reflect & Discuss

4. Assume that we ascribe to science the aim of describing the true nature of reality.

- **Would an empiricist accept or reject this idea without qualification?**
- **How is the religious agnostic similar to the scientific empiricist?**

(religious agnosticism: from Greek *agnōstos* for ‘unknowable’, i.e., the view that it is impossible to know or to say definitely whether or not there is a God).)

Underdetermination

- Grant science its aims of producing **true** accounts (descriptions) of both the observable causes and constituents (that's the *realist* position).

2. The Position of the Empiricist:

- The empiricist will do this but will also point out the problems of **underdetermination**:
 - **All confirmation**, that is, all distinctions of the *likely-to-be-true* from the *likely-to-be-false* theories, **will be seriously equivocal**, and testing against the evidence cannot distinguish any one of the unlimited number of empirically equivalent theories.
 - In addition, there is a problem with explanation: **Explanation** in general, judged by its contribution to our understanding of things, is a pragmatic and psychological event with **no clearly truth-conducive features**.
 - If we specialize in strictly causal explanation, the evidence still underdetermines which causal explanation is the right causal explanation.

Underd

- To paraphrase
 - Nancy Cartwright



“If God tells us (that is, we are certain) that nuclear fusion *explains* the sun’s energy, we can accept this and yet have no reason to believe that nuclear fusion is an actual solar process.

But if God tells us that nuclear fusion *causes* the sun’s energy, then we know as well that the nuclear fusion really takes place.”

Underde

- To paraphrase
 - Nancy Cartwright



“This is good news as far as it goes, but of course God doesn’t go around saying these things, nor does the natural evidence speak in such unequivocal terms of the correct causal explanation.”

Underdetermination

- Grant science its aims of producing **true** accounts (descriptions) of both the observable causes and constituents (that's the *realist* position).

2. The Position of the Empiricist:

- demands *agnosticism* towards unobservables.
- The idea is not to say that there are no such things as electrons or quarks. It is rather to say that we simply don't know and we in fact *can't* know if there are such things.
 - A claim about electrons or quarks is beyond the limits of justification and so beyond the limits of knowledge – the responsible attitude is to withhold judgement.
 - Though there is a good reason to use the theory **as if it is true**, there is no good reason to *believe that it is true*.